CASE STUDY · FRAUD DETECTION MODELING

Predicting Retailer Fraud Before It Becomes a Six-Figure Loss

A validated machine learning model built to catch a rare, high-severity fraud pattern before it turns into a catastrophic loss.

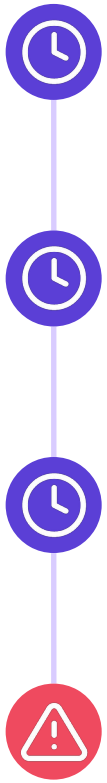


THE CHALLENGE

A Loss That Builds Quietly, Then Hits All at Once

A small number of retailers exploited inventory credit terms in order to gamble with their own product, hoping to secure a big prize. The retailers cashed smaller prize tickets in order to pay their weekly invoice. At a certain point, the invoice became too large to manage, and they couldn't pay their weekly bill. This ended up becoming a large, often six-figure loss and an inventory write-off.

One of the challenges in detecting and mitigating these instances is that the activity appears normal. Gambling and fraud instances are hard to detect unless an analyst is trained in reviewing certain metrics, and the importance of those metrics alone is difficult to quantify.



Receive Inventory

Retailer takes on instant ticket inventory, usually on credit

Play Own Product

Validates and cashes tickets instead of selling, hoping to secure a big prize

Debt Builds Quietly

Balance grows while activity looks normal

Settlement Hits

Invoice comes due that can't be paid

THE APPROACH

Machine Learning Model Built on Past Instances



Tight Scope

Predicts one specific, severe outcome within a 120-day window.



Multi-Source Signal

Combines internal sales and inventory activity with third-party credit data and redemption patterns.



Rare-Event Rigor

Upsampled for training, then tested on held-out live cases the model never saw.

HOW IT WORKS

Model Workflow



Data Sources

Sales, inventory, redemption history & business credit signals



XGBoost Model

Gradient-boosted classifier scores risk of a severe loss in 120 days



Risk Flag

High-risk accounts flagged for analyst review and analytical deep dive

THE RESULT

What the Model Delivered

100%

Confirmed fraud cases (9 of 9) correctly flagged in testing

~85%

Reduction in this loss pattern since the model went live

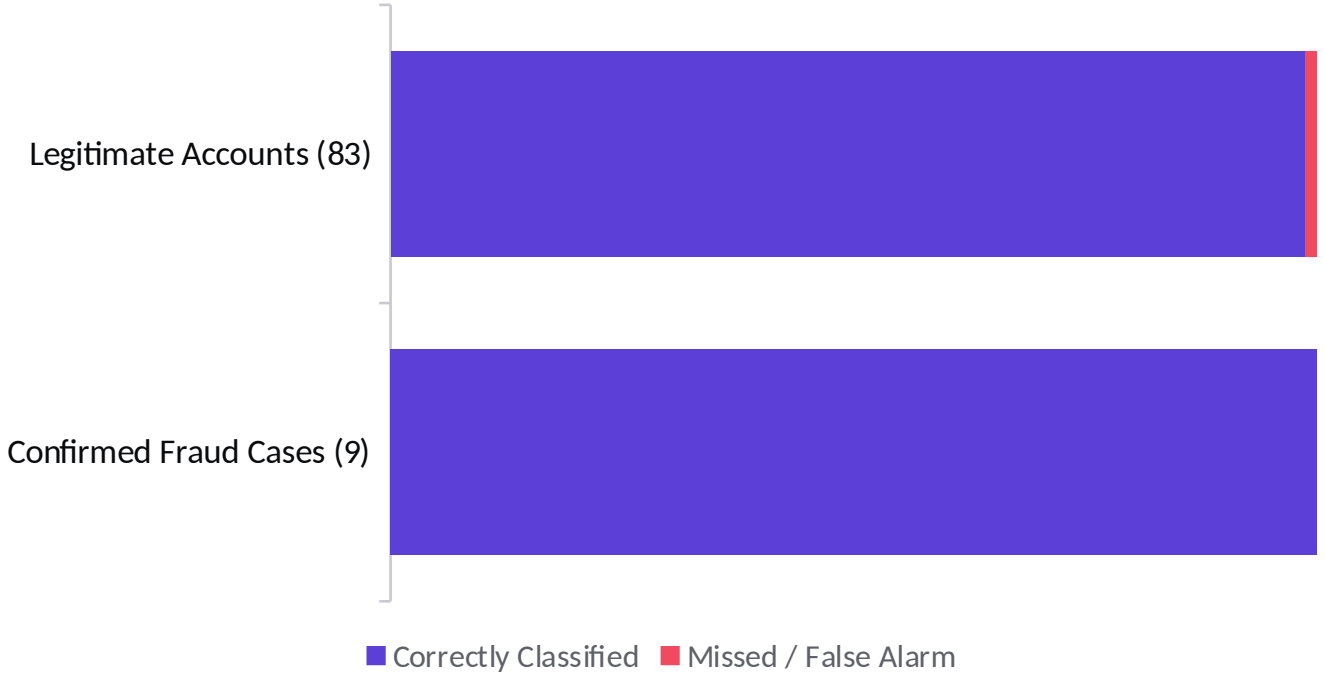
$2.1M

Additional exposure preventable over the next 5 years

Reduction measured against this loss pattern’s pre-model annual average. Five-year projection assumes current detection rates hold.

TEST RESULTS

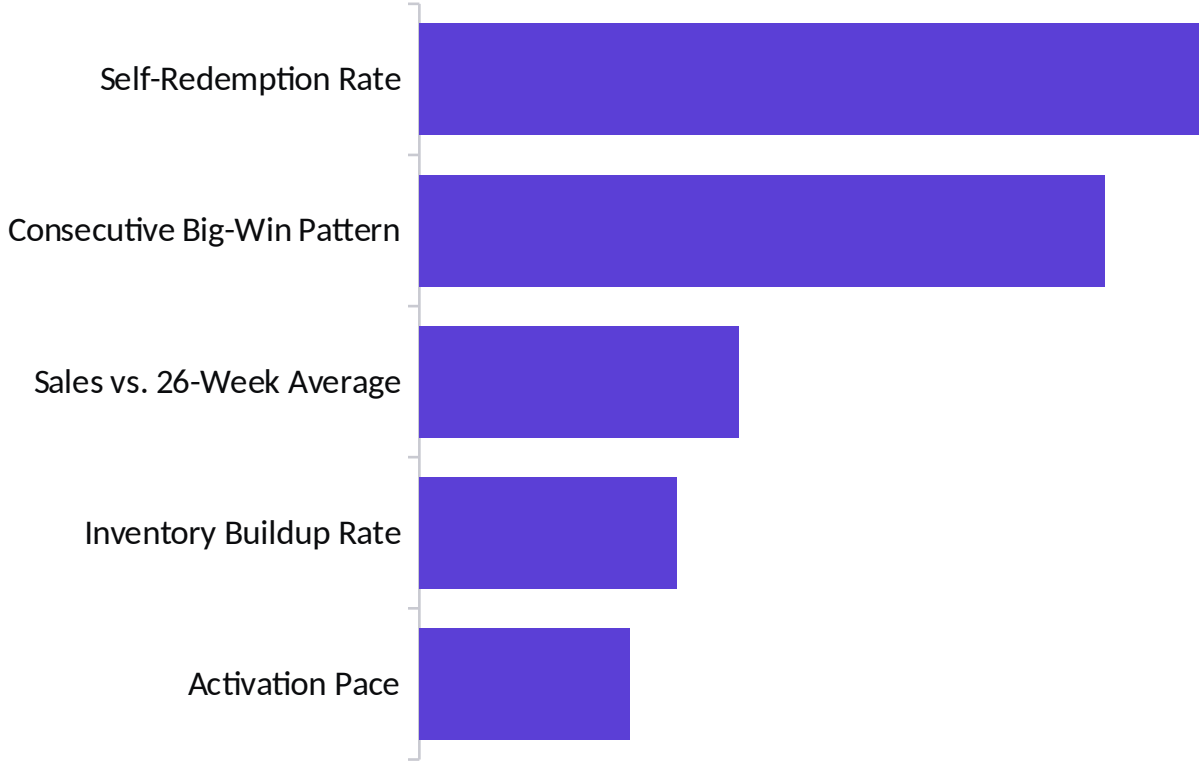
Model Performance on Held-Out Test Data



On a held-out test set the model never trained on, it flagged all 9 confirmed fraud cases and raised only one false alarm among 83 legitimate accounts.

WHAT THE MODEL LEARNED

Variable Importance



Retailers potentially committing fraud show several tell-tale signs: cashing an unusually high share of their own instant ticket inventory, and validating winning tickets in large, suspicious blocks. These patterns are almost never seen in ordinary consumer purchasing.

Sales trends, inventory buildup, and credit signals also contributed, but their relative weight would have been difficult to quantify without advanced modeling.

WHY IT MATTERS

Early Detection Prevents Losses and Saves Retailers

Small Losses Don’t Stay Small

Left undetected, fraud instances can develop into large, unmanageable losses in the hundreds of thousands.

Early Detection Can Save the Retailer

With early detection, analysts can apply prevention strategies to curb risk and keep the retailer from incurring a large loss, potentially salvaging the Lottery–retailer relationship.